

DUNMAN HIGH SCHOOL
2023 H2 PHYSICS (YEAR 6)

7	(a)	92 protons and 92 electrons 143 neutrons (in the nucleus)	B1 B1
	(b)(i)	α -particle travels short distance in air	B1
		<i>Marker's comments: Majority of candidates did not state the effect of interaction of alpha particles with air.</i>	
	(b)(ii)	very small proportion in backwards direction / large angles majority pass through with no /small deflections either most of mass is in very small volume (nucleus) and is charged or most of atom is empty space	B1 B1 B1
		<i>Marker's comments: Qualitative answers were poorly presented to explain the observations.</i>	
	(c)	$I = Q / t = nq / t$ $n / t = (1.5 \times 10^{-12}) / (2 \times 1.60 \times 10^{-19})$ $n / t = 4.7 \times 10^6 \text{ s}^{-1}$	C1 A1
		<i>Marker's comments: Power of ten was wrongly quoted.</i>	